

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5					
	A.	C	B	B	D	C					
SECTION-I(ii)	Q.	6	7	8	9	10	11	12	13	14	15
	A.	B,C	B,C	A,B,C	A,B,C	A,C,D	A,D	A,B,C,D	A,C	B,C,D	A,C
SECTION-II	Q.	1	2	3	4	5					
	A.	2.00	3.00	45.00	8.00	5.00					

PART-2 : CHEMISTRY

SECTION-I(i)	Q.	1	2	3	4	5					
	A.	C	B	A	A	B					
SECTION-I(ii)	Q.	6	7	8	9	10	11	12	13	14	15
	A.	A,C,D	A,B,C,D	A,B,C	A,B	A,B,D	A,B,C	A,B,C,D	A,B,C,D	A,B,C	A,D
SECTION-II	Q.	1	2	3	4	5					
	A.	149.00	16.00	2.00	5.00	150.00					

PART-3 : MATHEMATICS

SECTION-I(i)	Q.	1	2	3	4	5					
	A.	C	C	A	A	A					
SECTION-I(ii)	Q.	6	7	8	9	10	11	12	13	14	15
	A.	B,C	A,B,C,D	A,B,C	A,B,C,D	A,B,C,D	A,C	A,C	A,C	A,B,C,D	A,C,D
SECTION-II	Q.	1	2	3	4	5					
	A.	20.21	2.66 or 2.67	13.00	5.00	2.44 or 2.45					

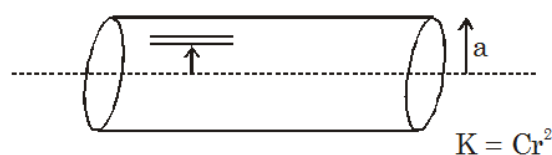
HINT – SHEET
PART-1 : PHYSICS
SECTION-I(i)

 1. **Ans (C)**

$$T V_1 \cos 0^\circ - 2T \sin(\theta) V_2 = 0$$

$$V_1 = 2V_2 \sin \theta$$

$$\frac{V_1}{V_2} = 2 \sin \theta$$

 2. **Ans (B)**


$$dR = \frac{L}{C r^2 2\pi r dr}$$

$$\frac{1}{R} = \int \frac{1}{dR} \int \frac{2\pi C r^3 dr}{L}$$

$$\frac{1}{R} = \frac{2\pi C}{L} \frac{a^4}{4}$$

$$\frac{R}{L} = \frac{2}{\pi C a^4}$$

 3. **Ans (B)**

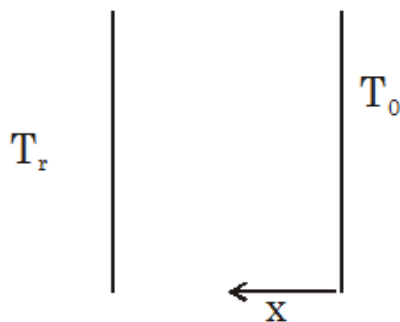
$$\frac{d}{D} = \frac{h}{f} \Rightarrow f = \frac{Dh}{d}$$

$$f = \frac{100 \times 1}{2} = 50 \text{ cm}$$

PART-1 : PHYSICS

SECTION-I(ii)

4. Ans (D)



$$i = \frac{(K_0 + \alpha x) A}{dx} dT$$

$$\int i \frac{dx}{K_0 + \alpha x} = \int A dT$$

for $x = 0$ to ℓ

$$\frac{1}{\alpha} \ln \left(\frac{K_0 + \alpha \ell}{K_0} \right) = A (T_0 - T_r)$$

for $x = 0$ to $\frac{\ell}{2}$

$$\frac{1}{\alpha} \ln \left(\frac{K_0 + \frac{\alpha \ell}{2}}{K_0} \right) = A (T_0 - T)$$

$$\frac{\ln 2}{\ln \left(\frac{3}{2} \right)} = \frac{25}{(45 - T)} \Rightarrow T = 30.5^\circ\text{C}$$

5. Ans (C)

$$h\nu - \phi = KE_{\max}$$

$$\frac{6.4 \times 10^{-34} \times 10^{15}}{1.6 \times 10^{-19}} \text{eV} - 2 = KE_{\max}$$

$$KE_{\max} = 2 \text{ eV}$$

6. Ans (B,C)

$$E_H = 13.6 \left(\frac{1}{1} - \frac{1}{9} \right) = 13.6 \times \frac{8}{9}$$

$$E_H - \phi = K_1 \dots (i)$$

$$\lambda_H = \frac{h}{\sqrt{2mK_1}} \dots (ii)$$

$$E = 13.6 Z^2 \left(1 - \frac{1}{4} \right) = 13.6 \times \frac{3}{4} \times Z^2$$

$$E - \phi = K_2 \dots (iii)$$

$$\frac{\lambda_H}{3} = \frac{h}{\sqrt{2mK_2}} \dots (iv)$$

$$\text{Also } 13.6 Z^2 \left(\frac{3}{4} \right) = 13.6 \times 3$$

$$z = 2$$

$$K_2 = 9K_1$$

$$13.6 \times 3 - \phi = 9 \left(13.6 \times \frac{8}{9} - \phi \right)$$

$$\phi = \frac{13.6 \times 5}{8} \text{eV} = 8.5 \text{eV}$$

8. Ans (A,B,C)

$$\frac{1 + n_B}{\frac{19}{13} - 1} = \frac{1}{\frac{5}{3} - 1} + \frac{n_B}{\frac{7}{5} - 1}$$

$$\frac{13}{6} (1 + n_B) = \frac{3}{2} + \frac{5n_B}{2}$$

$$\Rightarrow n_B = 2$$

$$V = \sqrt{\frac{\gamma RT}{M}}$$

$$M = \frac{1 \times 4 + 2 \times 32}{3}$$

$$M = \frac{68}{3}$$

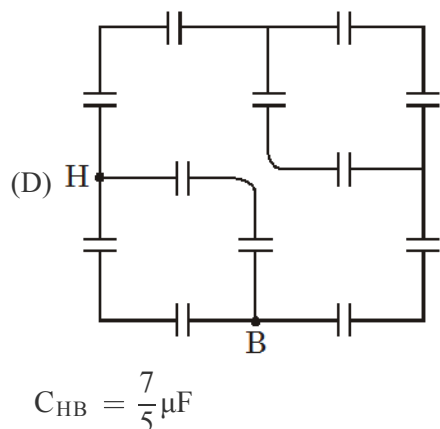
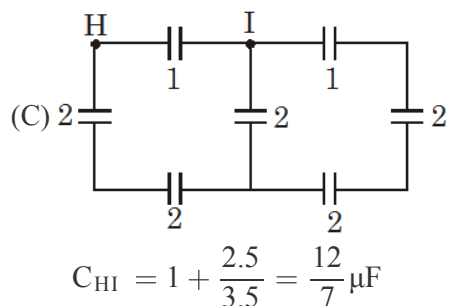
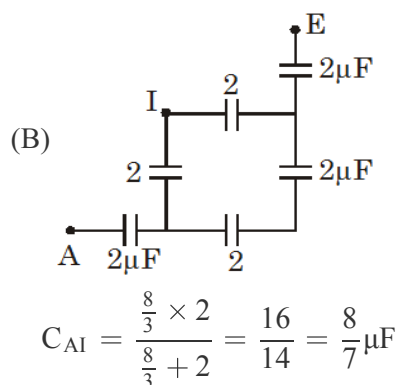
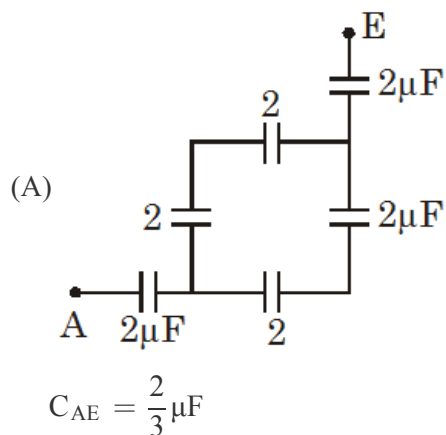
$$V = \sqrt{\frac{19}{13} \times \frac{25}{3} \times \frac{300 \times 3}{68 \times 10^{-3}}}$$

$$V = 401 \text{ m/s}$$

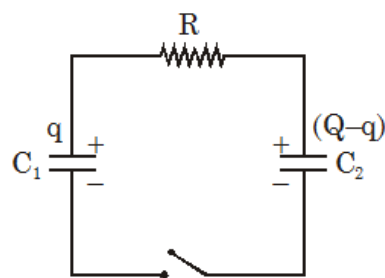
$$\frac{\Delta V}{V} \times 100 = \frac{1}{2} \frac{\Delta T}{T} \times 100$$

$$= 50 \times \frac{1}{300} = \frac{1}{6}$$

9. Ans (A,B,C)



10. Ans (A,C,D)



$$\frac{q}{C} - iR - \frac{(Q-q)}{C} = 0 \quad \& \quad i = -\frac{dq}{dt}$$

$$-iR = \frac{Q-2q}{C}$$

$$q = \frac{Q}{2} \left(1 + e^{-\frac{2t}{RC}} \right)$$

$$i = \frac{Q}{RC} e^{-\frac{2t}{RC}}$$

$$\Delta U = \frac{\left(\frac{Q}{2} \right)^2}{2C} \times 2 - \frac{Q^2}{2C} = -\frac{Q^2}{4C}$$

11. Ans (A,D)

$$E4\pi r^2 = \frac{Q + \int_R^r \rho 4\pi r^2 dr}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2} + \frac{\beta 4\pi}{4\pi\epsilon_0 r^2} \times \frac{(r^2 - R^2)}{2}$$

$$Q = 2\pi\beta R^2, \quad E = \frac{\beta}{2\epsilon_0}$$

12. Ans (A,B,C,D)

(A) $\Delta x_0 = (d \sin \phi) \times 1.2 - (2.4 - 1.4)t$
 $= 0.6 \text{ mm} - 0.6 \text{ mm} = 0$

(B) $\Delta x_{CM} = d \sin \phi \times 1.2 - d \sin \theta \times 1.4 = 0$

$$\sin \theta = \frac{6}{14} \Rightarrow \theta = \sin^{-1} \left(\frac{3}{7} \right)$$

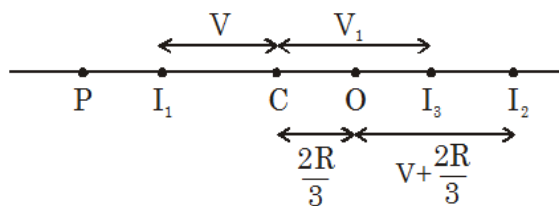
(C) $\beta = \frac{D\lambda}{d\mu} = \frac{5 \times 10^{-7}}{10^{-3} \times 1.4} = \frac{5}{14} \text{ mm}$

(D) $\Delta x_0 = 0 \Rightarrow I = 4I'$

SECTION-II

13. Ans (A,C)

$$\frac{\mu}{V} - \frac{3}{R} \Rightarrow V = \frac{2\mu R}{3}$$



$$\frac{3}{V_1} - \frac{\mu}{V + \frac{4R}{3}} = 0 \Rightarrow V_1 = \frac{\mu R + 2R}{\mu}$$

Refraction from curved surface

$$\frac{\frac{5}{2}}{-\frac{5R}{3}} + \frac{\frac{3}{2}}{\frac{2R(\mu+1)}{\mu}} = \frac{\frac{5}{2} - \frac{3}{2}}{-R}$$

$$-\frac{3}{2R} + \frac{3\mu}{4R(\mu+1)} = \frac{1}{-R}$$

$$3\mu = 2\mu + 2 \Rightarrow \mu = 2$$

$$d_{I_1 I_2} = 4R = 80\text{cm}$$

14. Ans (B,C,D)

$$F = -\frac{du}{dx} = -10x + 20$$

$v_{\max}$ occurs at equilibrium position.

$$\Rightarrow x = 2$$

$$T = 2\pi\sqrt{\frac{0.1}{10}} = \frac{\pi}{5}\text{sec}$$

15. Ans (A,C)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \frac{v}{u} = \pm 4$$

$$\frac{1}{4v} + \frac{1}{v} = \frac{1}{-25} \quad -\frac{1}{4u} + \frac{1}{u} = \frac{1}{-25}$$

$$v = -\frac{125}{4} \quad u = -\frac{75}{4}$$

1. Ans (2.00)

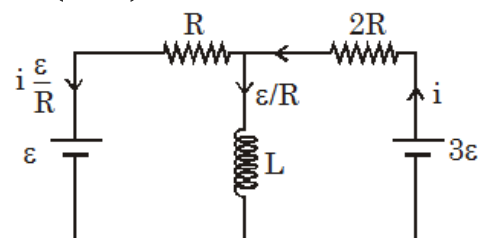
$$\frac{A_1}{A_2} \frac{e^{-\lambda_1 t}}{e^{-\lambda_2 t}} = \frac{20}{80}$$

$$\frac{A_1}{A_2} = \frac{80}{20}$$

$$e^{(\lambda_2 - \lambda_1)t} = \frac{1}{16}$$

$$t = \frac{1}{\lambda_1 - \lambda_2} \ln 16 = 18 \ln 16$$

2. Ans (3.00)



$$3\epsilon - i2R - R\left(i - \frac{\epsilon}{R}\right) - \epsilon = 0$$

$$i = \frac{\epsilon}{R}$$

$$\Delta V_{BE} = \epsilon = \frac{L di}{dt} \Rightarrow \left| \frac{di}{dt} \right| = 10$$

3. Ans (45.00)

$$P_i + \rho g 15 = P_{\text{atm}} = \rho g 75$$

$$P_i = \rho g 60$$

$$P_f = P_{\text{atm}} + \rho g 15 = \rho g 90$$

$$\frac{P_f (A \times 40)}{T_f} = \frac{P_i (A \times 40)}{300}$$

$$T_f = \frac{3}{2} \times 300 = 450\text{K}$$

4. Ans (8.00)

$$t = \sqrt{\frac{2 \times 20}{10}} = 2\text{sec}$$

$$v = \frac{80}{2} = 40$$

$$P_0 + \rho g 53 = 10^5 + \frac{1}{2} \rho v^2$$

$$P_0 = 10^5 + 660 \times 270$$

$$= 2.78 \times 10^5$$

$$a + b = 8$$

5. Ans (5.00)

$$I = \frac{P}{4\pi r^2} = \frac{(\Delta p)^2}{2\rho V}$$

$$\Delta p = 5$$

PART-2 : CHEMISTRY

SECTION-II

5. Ans (150.00)

$$P_t = \sqrt{P_A^0 \cdot P_B^0}$$

$$= 150.00 \text{ torr}$$

PART-3 : MATHEMATICS

SECTION-I(i)

1. Ans (C)

Any point on circle be $(\cos\theta, \sin\theta)$

If this point lies on the curve $\sin x = \cos y$

$$\Rightarrow \sin(\cos\theta) = \cos(\sin\theta)$$

$$\Rightarrow \frac{\pi}{2} - \cos\theta = 2n\pi \pm \sin\theta$$

$$\Rightarrow \cos\theta \pm \sin\theta = \frac{\pi}{2} - 2n\pi$$

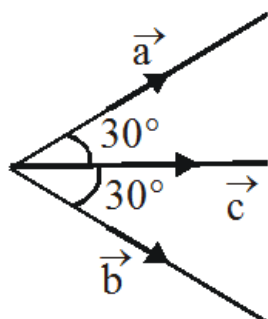
$$\text{but } \left| \frac{\pi}{2} - 2n\pi \right| > \sqrt{2} \forall n \in \mathbb{I} \Rightarrow \text{No solution.}$$

$\therefore$ Number of points of intersection = 0.

2. Ans (C)

$$a^2 + b^2 + c^2 + 2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a}$$

$$= 14 + 2 \times \left(2 \times \frac{1}{2} + 6 \times \frac{\sqrt{3}}{2} + \frac{3\sqrt{3}}{2} \right)$$



$$= 14 + 2 + 6\sqrt{3} + 3\sqrt{3} = 16 + 9\sqrt{3}$$

3. Ans (A)

$$\frac{d}{dx} (y(f(x) + f'(x))) = 1 \Rightarrow y \left(y + \frac{dy}{dx} \right) = x + c$$

$$\Rightarrow y \frac{dy}{dx} + y^2 = x + c$$

$$\text{Let } y^2 = t$$

$$\Rightarrow \frac{dt}{dx} + 2t = 2x + 2c$$

$$\text{I. F} = e^{\int 2dx} = e^{2x}$$

$$\therefore t \cdot e^{2x} = \int 2(x+c) e^{2x} dx$$

where c is integration constant.

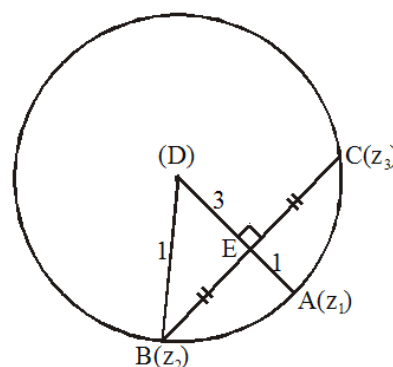
$$\Rightarrow t \cdot e^{2x} = 2(x+c) \frac{e^{2x}}{2} - \int e^{2x} dx$$

$$\Rightarrow t \cdot e^{2x} = (x+c) e^{2x} - \frac{e^{2x}}{2} + k$$

$$\Rightarrow t = x + c - \frac{1}{2} + k_1 e^{-2x}$$

$$\Rightarrow y^2 = k_1 e^{-2x} + x + k_2$$

4. Ans (A)



A, B, C lies on unit circle centered at origin.

$$\text{Also } \frac{3z_1 + 1.0}{4} = \frac{z_2 + z_3}{2}$$

Let $DE = 3k$ & $EA = k$

$$4k = 1 \Rightarrow k = \frac{1}{4}$$

$$BE = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4} \Rightarrow |z_2 - z_3| = BC = \frac{\sqrt{7}}{2}$$

5. Ans (A)

Total points generated = $4 \times 4 \times 4 = 64$

Three points can be selected ${}^{64}C_3$

$$x_1 < x_2 < x_3 \text{ \& } \frac{x_1 + x_2 + x_3}{3} \in I$$

allows ordered pair (x_1, x_2, x_3)

$$(0, 1, 2), (1, 2, 3) \in I$$

allows ordered pair (z_1, z_2, z_3)

$$(0, 0, 0), (1, 1, 1), (3, 0, 0), (2, 1, 0), (2, 2, 2)$$

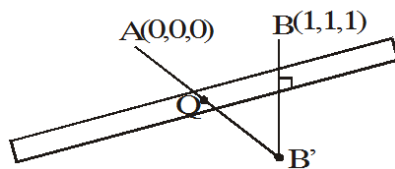
$$(0, 3, 3), (3, 3, 3), (3, 2, 1) \rightarrow 8 \text{ ways.}$$

allows ordered pair $(y_1, y_2, y_3) \rightarrow 8 \text{ ways.}$

$$\text{Required probability} = \frac{2 \times 8 \times 8}{{}^{64}C_3}$$

SECTION-I(ii)

6. Ans (B,C)



For $(QA + QB)$ min, Q must lie on AB' line

where B' is mirror image of B in plane $x - y + z = 3$

$$B' \equiv \left(\frac{7}{3}, \frac{-1}{3}, \frac{7}{3} \right)$$

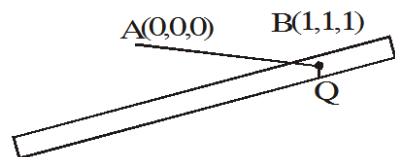
Q is intersection of AB' and plane

$$Q \left(\frac{7}{5}, \frac{-1}{5}, \frac{7}{5} \right) \equiv (a, b, c)$$

$$\text{Hence } a - c - 10b = 2$$

For $|QA - QB|_{\min}$, Q must lie on point of

intersection of AB with plane



$$Q \equiv (3, 3, 3) \equiv (a, b, c)$$

$$\text{Hence } a - 2b + c = 0$$

7. Ans (A,B,C,D)

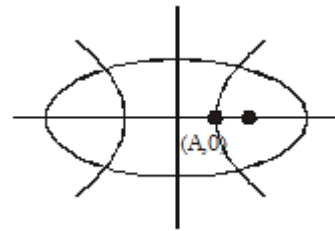
$$\frac{{}^{15}C_5 (x^3)^{10} (\sqrt{2} \cdot x^{-2})^5}{{}^{15}C_5 (\sqrt{2}x^{-2})^{10} (x^3)^5} = \frac{4\sqrt{2}}{243}$$

$$\Rightarrow \left(\frac{x^3}{\sqrt{2}x^{-2}} \right)^5 = \left(\frac{\sqrt{2}}{3} \right)^5$$

$$\frac{x^5}{\sqrt{2}} = \frac{\sqrt{2}}{3} \Rightarrow x^5 = \frac{2}{3}$$

$$x = \left(\frac{2}{3} \right)^{\frac{1}{5}}$$

8. Ans (A,B,C)



Hyperbola and ellipse will be confocal with focus $(\pm 2\sqrt{2}, 0)$

$$1 - e_1^2 = \frac{1}{9} \Rightarrow e_1 = \frac{2\sqrt{2}}{3} \Rightarrow e_2 = \frac{3}{2\sqrt{2}}$$

$$\therefore \frac{x^2}{A^2} - \frac{y^2}{B^2} = 1 \Rightarrow A = \frac{8}{3}$$

$$B^2 = \frac{64}{9} \left(\frac{9}{8} - 1 \right) \Rightarrow B^2 = \frac{8}{9}$$

$$\frac{9x^2}{64} - \frac{9y^2}{8} = 1$$

9. Ans (A,B,C,D)

$$(A) \quad P(\bar{D}) = \frac{90}{100} \times \frac{10}{99} = \frac{1}{11}$$

(B) Required probability

$$= \frac{{}^{99}C_{86}}{{}^{100}C_{86}} \times \frac{1}{14} = \frac{1}{100}$$

(C) E : 2nd article is defective

A : 1st article is good

B : 1st article is defective

$$P\left(\frac{A}{E}\right) = \frac{P\left(\frac{E}{A}\right)P(A)}{P\left(\frac{E}{A}\right)P(A) + P\left(\frac{E}{B}\right)P(B)}$$

$$= \frac{10}{99} \times \frac{90}{100} = \frac{\frac{10}{99} \times \frac{90}{100}}{\frac{10}{99} \times \frac{90}{100} + \frac{10}{100} \times \frac{90}{99}} = \frac{10}{11}$$

Hence option 'C' is correct

(D) is also correct.

10. Ans (A,B,C,D)

The number of natural numbers in r^{th} group is r^3 .

So the total number of natural numbers in first

$$n \text{ group} = \sum_{r=1}^n r^3 = \left(\frac{n(n+1)}{2} \right)^2$$

The n^{th} group therefore consists of the numbers

$$\left(\frac{1}{4}(n-1)^2(n)^2 + 1, \frac{1}{4}(n-1)^2n^2 + 2, \dots, \frac{1}{4}(n-1)^2n^2 + n^3 \right)$$

Sum of numbers in n^{th} group is

$$\begin{aligned} n^3 \cdot \frac{1}{4}(n-1)^2n^2 + \frac{1}{2}n^3(n^3 + 1) \\ = \frac{n^3}{4} [n^2(n-1)^2 + 2n^3 + 2] = \frac{n^3}{4} (n^4 + n^2 + 2) \end{aligned}$$

11. Ans (A,C)

Let $f(x) = ax^2 + c$

$$f(1) = a + c \quad -4 \leq a + c \leq -1$$

$$f(2) = 4a + c \quad -1 \leq 4a + c \leq 5$$

$$\Rightarrow a = \frac{f(2) - f(1)}{3}, f(0) = c \Rightarrow \frac{4f(1) - f(2)}{3}$$

$$f(3) = 9a + c$$

$$= 3f(2) - 3f(1) +$$

$$\frac{4f(1)}{3} - \frac{f(2)}{3} = \frac{8f(2) - 5f(1)}{3}$$

$$f(3)_{\max} = \frac{8.5 - 5(-4)}{3} = 20$$

12. Ans (A,C)

$$\angle B = \sec^{-1} \left(\frac{5}{4} \right) + \operatorname{cosec}^{-1} \sqrt{5}$$

$$= \tan^{-1} \left(\frac{3}{4} \right) + \tan^{-1} \left(\frac{1}{2} \right) = \tan^{-1} \frac{\frac{3}{4} + \frac{1}{2}}{1 - \frac{3}{4} \cdot \frac{1}{2}} = \tan^{-1} 2$$

$$\angle C = \operatorname{cosec}^{-1} \left(\frac{25}{7} \right) + \cot^{-1} \left(\frac{9}{13} \right)$$

$$= \tan^{-1} \left(\frac{7}{24} \right) + \tan^{-1} \left(\frac{13}{9} \right)$$

$$= \tan^{-1} \left(\frac{\frac{7}{24} + \frac{13}{9}}{1 - \frac{7}{24} \cdot \frac{13}{9}} \right) = \tan^{-1} 3$$

$$\tan B = 2 \text{ \& } \tan C = 3 \Rightarrow \tan A = 1$$

$$\therefore \sum \tan A = \prod \tan A$$

$$\therefore \sin A = \frac{1}{\sqrt{2}}, \sin B = \frac{2}{\sqrt{5}} \text{ and } \sin C = \frac{3}{\sqrt{10}}$$

$$\therefore \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\Rightarrow \sqrt{2}a = \frac{\sqrt{5}b}{2} = \frac{3\sqrt{10}}{3}$$

$$\text{Hence } a = \sqrt{5} \text{ and } b = 2\sqrt{2}, c = 3$$

$$\tan A = 1, \tan B = 2, \tan C = 3 \text{ are in A.P.}$$

The triangle with sides a^2 , $b^{\frac{4}{3}}$ and c will have

side-length 5, 4 and 3 respectively

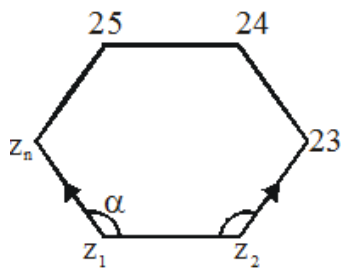
$\therefore$ Distance between orthocentre and centroid

$$= \frac{2}{3} (\text{circumradius}) = \frac{\text{hypotenuse}}{3} = \frac{5}{3} \text{ Ans.}$$

13. Ans (A, C)

$$\begin{aligned}
 P &= \frac{1}{n^4} \prod_{r=1}^{2n} (n^2 + r^2)^{\frac{1}{n}} \\
 &= \prod_{r=1}^{2n} \left(\frac{n^2 + r^2}{n^4} \right)^{\frac{1}{n}} = \left(\frac{n^2 + 1^2}{n^2} \right)^{\frac{1}{n}} \cdot \left(\frac{n^2 + 2^2}{n^2} \right)^{\frac{1}{n}} \\
 &\quad \dots \left(\frac{n^2 + (2n)^2}{n^2} \right)^{\frac{1}{n}} \\
 &= \left(1 + \left(\frac{1}{n} \right)^2 \right)^{\frac{1}{n}} \left(1 + \left(\frac{2}{n} \right)^2 \right)^{\frac{1}{n}} \left(1 + \left(\frac{3}{n} \right)^2 \right)^{\frac{1}{n}} \\
 &\quad \dots \left(1 + \left(\frac{2n}{n} \right)^2 \right)^{\frac{1}{n}} \\
 \ln P &= \frac{1}{n} \sum \ln \left(1 + \frac{r^2}{n^2} \right) \\
 &= \int_0^2 \ln(1+x^2) dx = \int_0^2 \ln(1+(2-x)^2) dx \\
 &= \int_0^2 \ln(x^2 - 4x + 5) dx
 \end{aligned}$$

14. Ans (A, B, C, D)



$$\begin{aligned}
 (z_2 - z_1)e^{i\alpha} &= z_n - z_1 \\
 (z_1 - z_2)e^{-i\alpha} &= z_3 - z_2 \\
 z_1(1 - e^{i\alpha} + e^{-i\alpha}) + z_2(1 + e^{i\alpha} - e^{-i\alpha}) &= z_n + z_3 \\
 A &= 1 + 2i \sin \alpha \\
 |A| &= \sqrt{1 + 4\sin^2 \frac{2\pi}{n}} \quad n = 8 \Rightarrow \sqrt{3} = |A| \\
 n = 4 \Rightarrow |A| &= \sqrt{5} \quad n = 12 \Rightarrow \sqrt{2} = |A| \\
 n = 6 \Rightarrow |A| &= 2
 \end{aligned}$$

15. Ans (A, C, D)

$\because f(x)$ and $g(x)$ are polynomials of degree 3.
then suppose $g(x) = \lambda(x - \alpha)^2(x - \beta)$ ($\alpha \neq \beta$)
and for existence of limit $f(x) = 0$, $f'(\alpha) = 0$, $f''(\alpha) = 0$.

$$\therefore f(x) = \mu(x - \alpha)^3.$$

$$\therefore f(x) \cdot g(x) = \lambda \cdot \mu \cdot (x - \alpha)^5(x - \beta)$$

$$\text{Also, } h(x) = f(x) \cdot g'(x) + g(x) \cdot f'(x) = \frac{d}{dx}$$

$$(f(x) \cdot g(x))$$

$$= \frac{d}{dx} (\lambda \mu (x - \alpha)^5 (x - \beta))$$

$$= \lambda \mu [(x - \alpha)^5 \cdot 1 + 5(x - \beta) \cdot (x - \alpha)^4]$$

$$= \lambda \mu (x - \alpha)^4 [(x - \alpha) + 5x - 5\beta]$$

$$h(x) = \lambda \mu (x - \alpha)^4 [6x - \alpha - 5\beta]$$

$\therefore h(x) = 0$ has two distinct roots $x = \alpha$ and

$$x = \frac{\alpha + 5\beta}{1 + 5}$$

which lies between α and β . Clearly $h'''(\alpha) = 0$.

SECTION-II

1. Ans (20.21)

$$\begin{aligned}
 k &= \int_{-2021}^{2021} \frac{1}{(1+x^{2019} + \sqrt{1+x^{4038}})} \cdot \frac{(1+x^{2019} - \sqrt{1+x^{4038}})}{(1+x^{2019} - \sqrt{1+x^{4038}})} dx \\
 &= \int_{-2021}^{2021} \frac{1+x^{2019} - \sqrt{1+x^{4038}}}{2 \cdot x^{2019}} dx \\
 &= \int_{-2021}^{2021} \underbrace{\frac{1}{2 \cdot x^{2019}}}_{\text{odd function}} dx + \int_{-2021}^{2021} \frac{1}{2} dx - \int_{-2021}^{2021} \underbrace{\frac{\sqrt{1+x^{4038}}}{x^{2021}}}_{\text{odd function}} dx \\
 &= 0 + 2021 + 0 = 2021 \\
 \therefore \frac{2021}{100} &= 20.21
 \end{aligned}$$

2. Ans (2.66 or 2.67)

$$f(x) = x^3 + x^2 + x + 1 = (x^2 + 1)(x + 1)$$

$$\int_{-2}^1 f(x)dx + \underbrace{\int_1^4 f^{-1}(y)dy + \int_0^1 f(x) - \int_0^1 f(x)dx}_{=0}$$

$$\Rightarrow 4 + \int_{-2}^0 f(x)dx \Rightarrow 4 + \left[\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} + x \right]_{-2}^0$$

$$= 4 - \left(4 - \frac{8}{3} + 2 - 2 \right) = \frac{8}{3}$$

3. Ans (13.00)

$$\text{Let } a = 5^{\alpha_1} \cdot 3^{\beta_1}$$

$$b = 5^{\alpha_2} \cdot 3^{\beta_2}$$

$$\text{and } c = 5^{\alpha_3} \cdot 3^{\beta_3}$$

α_1	α_2	α_3
2	0	3
2	1	3
2	2	3
0	2	3
1	2	3

$\alpha_1, \alpha_2, \alpha_3$ can be selected in 5 ways

where $\beta_1, \beta_2, \beta_3$ can be selected in

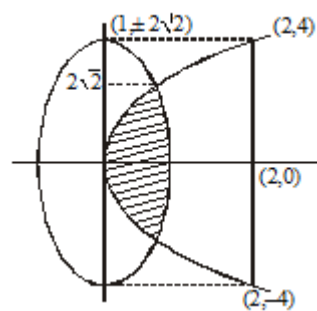
13 ways

$$\therefore \text{Total ways} = 5 \times 13 = 65$$

$$\Rightarrow \frac{k}{5} = 13$$

4. Ans (5.00)

Solving parabola & ellipse we get



$$\text{Sample space } s = 2 \int_0^2 2\sqrt{2x} = \frac{32}{3}$$

for area

$$= 2 \int_0^{2\sqrt{2}} \left(\sqrt{2} \sqrt{1 - \frac{y^2}{16}} - \frac{y^2}{8} \right) dy = \frac{\sqrt{2}(3\pi + 2)}{3}$$

$$\text{Probability} : \frac{\sqrt{2}(3\pi + 2)}{32} \Rightarrow a + b = 5$$

5. Ans (2.44 or 2.45)

$$\vec{P} = \frac{\vec{a}}{2}, \vec{Q} = \frac{\vec{b}}{2}, \vec{R} = \frac{\vec{c}}{2}, \vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{4}$$

equation of

$$\overrightarrow{GR} \Rightarrow \vec{r} = \frac{\vec{c}}{2} + \lambda \left(\frac{\vec{a} + \vec{b} + \vec{c}}{4} - \frac{\vec{c}}{2} \right)$$

$$\text{equation of } \overrightarrow{PQ} \Rightarrow \vec{r} = \frac{\vec{a}}{2} + \mu \left(\frac{\vec{b}}{2} - \frac{\vec{a}}{2} \right)$$

$$\text{S.D.} = \frac{\left| \left(\frac{\vec{a}}{2} - \frac{\vec{c}}{2} \right) \cdot (\vec{p} \times \vec{q}) \right|}{|\vec{p} \times \vec{q}|} = \frac{1}{2\sqrt{6}};$$

where $\vec{a} = \hat{i}, \vec{b} = \hat{j}, \vec{c} = \hat{k}$

$$\vec{p} = \left(\frac{\vec{a} + \vec{b} + \vec{c}}{4} - \frac{\vec{c}}{2} \right), \vec{q} = \left(\frac{\vec{b}}{2} - \frac{\vec{a}}{2} \right)$$

JEE(Main+Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2.25	648.00	1.14	11.42	72.00	96.00	2.54	2.47	240.59	0.14
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	1	8	4	2	2	2	3	2	2

PART-2 : CHEMISTRY

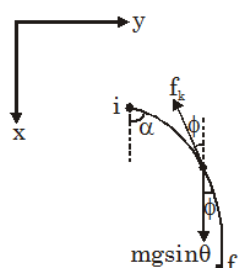
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	16.00	27.00	54.44	2.00	4.00	7.00	10.00	41.50	0.46	2.50
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	8	2	6	4	6	6	8	6	5

PART-3 : MATHEMATICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	42.66 or 42.67	2.00	2.57	3.00	36.00	0.70 or 0.71	20.00	3280.00	106.00	3.33 or 3.34
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	2	0	1	9	4	1	8	7	6

HINT – SHEET
PART-1 : PHYSICS
SECTION-II
1. Ans (2.25)

f_r must be zero because the block's final speed and hence its final centripetal acceleration must approach zero, which can only happen if the frictional force points up the plane, opposite the weight component down the plane.



$$\frac{dv_x}{dt} = g \sin \theta - \mu_k g \cos \theta \cos \phi \quad \dots(1)$$

$$\frac{dv}{dt} = g \sin \theta \cos \phi - \mu_k g \cos \theta \dots(2)$$

$$\text{Eliminate } \cos \phi. \mu_k \frac{dv}{dt} = g \cos \theta (\tan^2 \theta - \mu_k^2) -$$

$$\tan \theta \frac{dv_x}{dt} \dots(3)$$

after simplifying. Now integrate both sides over the time t from the initial to the final point to obtain

$$\mu_k (0 - v_0)$$

$$= g \cos \theta (\tan^2 \theta - \mu_k^2) t - \tan \theta (0 - v_0 \cos \alpha)$$

.....(4)

$$\text{which rearranges as } t = \frac{v_0 (\mu_k + \tan \theta \cos \alpha)}{g \cos \theta (\mu_k^2 - \tan^2 \theta)} \dots\dots$$

(5)

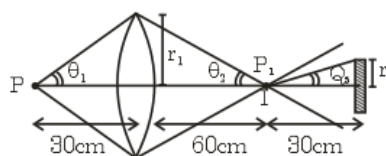
2. Ans (648.00)

$$\tau = F \times 2r$$

$$\alpha = \frac{2Fr}{I}$$

$$\text{Work done} = \tau \theta$$

$$= \tau \times \frac{1}{2} \propto t^2 = 648 \text{ J}$$

3. Ans (1.14)


Power incident on lens

$$\rho_1 = \frac{P}{4\pi} 2\pi (1 - \cos \theta_1)$$

Power incident on Area

$$= \frac{P_1}{2\pi(1 - \cos \theta_2)} 2\pi(1 - \cos \theta_3)$$

$$P_2 = \frac{P}{2} \frac{(1 - \cos \theta_1)(1 - \cos \theta_2)}{(1 - \cos \theta_2)}$$

$$P_2 = \frac{P}{2} \frac{2\sin^2\left(\frac{\theta_1}{2}\right) \cdot 2\sin^2\left(\frac{\theta_3}{2}\right)}{2\sin^2\left(\frac{\theta_2}{2}\right)}$$

∴ Angle are very small

$$P_2 = P \frac{\theta_1^2 \cdot \theta_3^2}{4Q_2^2} = \frac{P \cdot \left(\frac{r_1}{30}\right)^2 \cdot \left(\frac{r_2}{30}\right)^2}{4\left(\frac{r_1}{60}\right)^2}$$

$$P_2 = \frac{4Pr_2^2}{4 \times (30)^2}$$

$$\pi r_2^2 = 0.5 \Rightarrow r_2^2 = \frac{1}{2\pi}$$

$$P_2 = \frac{2P}{3600\pi}$$

No. of photons incident

$$= \frac{P_2}{(hc/\lambda)} = \frac{P_2 \lambda}{hc}$$

$$N = \frac{2P}{3600\pi} \times \frac{4000}{12420}$$

No. of e⁻ emitted per unit time

$$n = \frac{20}{100} N$$

$$n = \frac{1}{5} \times \frac{2P \times 4000}{3600\pi \times 12420 \text{ eV}}$$

i = charge emitted per unit time

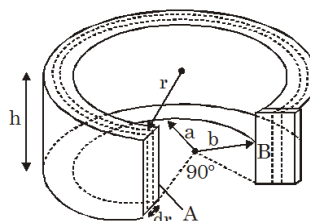
$$i = ne$$

$$i = \frac{1}{5} \times \frac{2P \times 4000e}{3600\pi \times 12420 \text{ eV}}$$

$$i = \frac{2P \times 4000}{5 \times 3600\pi \times 12420} \text{ A}$$

$$i = 1.14 \text{ mA}$$

4. Ans (11.42)



$$\text{Resistance of strip} = \frac{\rho \left(\frac{3\pi r}{2}\right)}{h dr} = \frac{3\pi r \rho}{2h dr}$$

All such strips are in parallel so overall resistance is given as

$$\frac{1}{R} = \int_a^b \frac{2h dr}{3\pi r \rho}$$

$$\frac{1}{R} = \frac{2h}{3\pi \rho} \ln \left(\frac{b}{a}\right)$$

$$R = \frac{3\pi \rho}{2h \ln 2} \text{ (as } b = 2a)$$

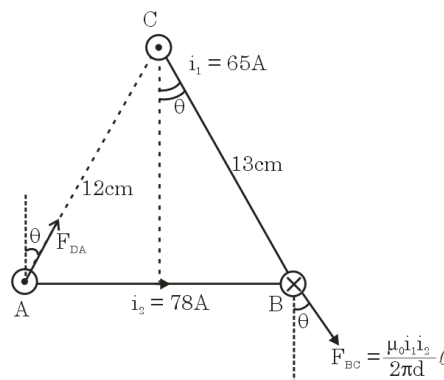
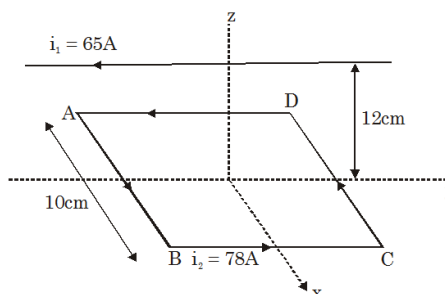
$$= \frac{3}{2} \times \frac{\pi \times 1.6 \times 10^{-8}}{10^{-2} \times 0.693}$$

$$= 11.42 \times 10^{-6}$$

$$= 11.42 \mu\Omega$$

5. Ans (72.00)

Side View



τ_{net} = Torque to force couple of $F \cos \theta$

$$= (F \cos \theta)(10 \text{ cm})$$

$$= \frac{\mu_0 i_1 i_2}{2\pi d} \ell \cos \theta. (10 \text{ cm})$$

Substituting the values we get

$$\tau = 72 \mu\text{N-m}$$

6. **Ans (96.00)**

$$\frac{A_i + A_r}{A_i - A_r} = \frac{3}{2}$$

$$\frac{A_i}{A_r} = 5 \dots (i)$$

$$\text{Now, } A_r^2 + A_t^2 = A_i^2$$

$$A_t^2 = A_i^2 - A_r^2$$

$$A_t = \sqrt{24} A_r$$

$$\text{Thus } \left(\frac{A_t}{A_i} \right)^2 = \frac{24}{25}$$

$$\frac{I_t}{I_i} = \frac{24}{25} \times 100\% = 96\%$$

8. **Ans (2.47)**

$$(S_1 O - t + \mu_1 t) - (S_2 O - t + \mu_2 t) = \Delta x$$

$$\Delta k = (\mu_1 - \mu_2)t$$

$$\Delta k = 0.1 \times 20 \times 10^{-6}$$

$$\Delta \phi = \frac{2\pi}{gl} \Delta x = \frac{2\pi}{4 \times 10^{-7}} \times 2 \times 10^{-6}$$

$$\Delta \phi = 10\pi$$

Max

$$I_{II} = \left(\sqrt{\frac{3}{4}} I_0 + \sqrt{\frac{I_0}{2}} \right)^2 \dots (i)$$

$$I_I = 4I_0 = 4 \text{ (given)}$$

$$\Rightarrow I_0 = 1 \dots (ii)$$

$$I_{II} = \left(\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \right)^2 I_0 = 2.47$$

9. **Ans (240.59)**

$$\text{Area} = \frac{\pi ab}{2} + b \times ae$$

$$\text{Time taken : } \frac{\frac{\pi ab}{2} + \frac{ab}{2}}{\pi ab} \frac{\frac{\pi}{2} + \frac{1}{2}}{\pi} \times 365 = 240.59$$

days

10. **Ans (0.14)**

Thickness of slip gauge

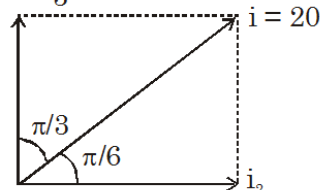
$$= 4.10 - 2.12 = 1.98 \text{ mm}$$

$$\text{Thus, error} = 2.12 - 1.98 = 0.14 \text{ mm}$$

SECTION-III

1. **Ans (6)**

$$i_1 = 20 \cos \frac{\pi}{3}$$



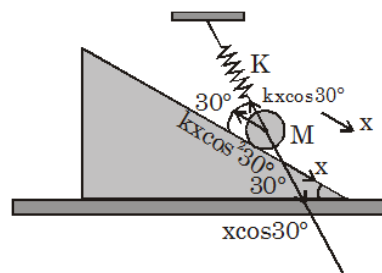
$$i_1 = 10 \sin \left(100\pi t + \frac{\pi}{3} \right)$$

$$V = 220\sqrt{2} \sin(100\pi t)$$

$$Z_1 = \frac{V_{rms}}{i_{rms}} = \frac{220}{\frac{10}{\sqrt{2}}}$$

$$Z_1 = 22\sqrt{2} \Omega$$

2. **Ans (1)**



$$F = -Kx \cos^2 30^\circ$$

$$ma = -(K \cos^2 30^\circ)x$$

$$a = -\left(\frac{K}{m} \frac{3}{4} \right) x$$

$$\omega = \sqrt{\frac{3K}{4m}}$$

$$\omega = \sqrt{\frac{3}{4m}} \times \frac{4m}{3} = 1 \text{ rad/sec}$$

3. **Ans (8)**

$$f_0 = \frac{v}{2\ell} = 200 \text{ Hz} \dots (i)$$

$$f_1 = \frac{v}{2\ell_1} \text{ \& } f_2 = \frac{v}{2\ell_2}$$

$$\text{Beats per sec} = f_1 - f_2 = \frac{v}{2} \left(\frac{1}{\ell_1} - \frac{1}{\ell_2} \right)$$

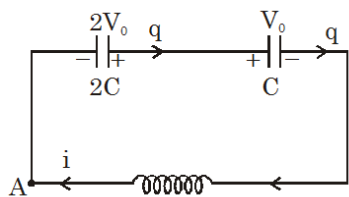
$$y = \frac{v}{2} \left(\frac{\ell_2 - \ell_1}{\ell_1 \ell_2} \right)$$

$$y = \frac{v}{2} \left(\frac{2}{49 \times 51} \right)$$

$$= 200 \times 100 \times \left(\frac{2}{49 \times 51} \right) = 16$$

$$\frac{16}{2} = 8$$

4. Ans (4)



$$\frac{(4CV_0 - q)}{2C} - \frac{CV_0 + q}{C} - L \frac{di}{dt} = 0$$

$$2V_0 - \frac{q}{2C} - V_0 - \frac{q}{C} = L \frac{di}{dt}$$

$$L \frac{d^2q}{dt^2} = V_0 - \frac{3q}{2C}$$

$$\frac{d^2q}{dt^2} = -\frac{3q}{2LC} + \frac{V_0}{L}$$

Solving above equation, we have

$$q = q_0 \sin(\omega t + \delta)$$

where q_0 is maximum value of q and $\omega = \sqrt{\frac{3}{2LC}}$

for maximum q

$$2V_0 - \frac{q}{2C} = V_0 + \frac{q}{C}$$

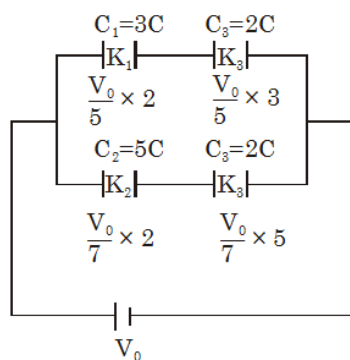
$$q = \frac{2CV_0}{3} = q_0$$

$$\text{Current, } i = \frac{dq}{dt}$$

$$i_{\max} = q_0 \omega$$

$$= \frac{2CV_0}{3} \sqrt{\frac{3}{2LC}} = 4A$$

5. Ans (2)



Ratio of breakdown voltages :

$$V_{B1} : V_{B2} : V_{B3} = 1 : 2 : 3$$

Hence C_1 will breakdown first.

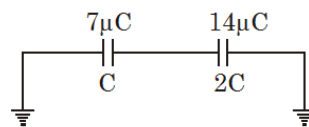
Electric field in C_1 ,

$$E_1 = \frac{\frac{2V_0}{5}}{\frac{d}{2}} = \frac{4V_0}{5d}$$

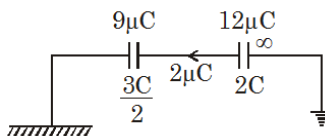
$$= 2 \times 10^3 \text{ V/m} = 2 \text{ V/mm}$$

6. Ans (2)

Initial circuit diagram



Final circuit diagram



$2\mu\text{C}$ charge will flow through thin wire

7. Ans (2)

Applying snell's law at each surface

$$\text{At B, } \sin i = n_1 \sin r_1$$

$$\Rightarrow \sin^2 i = n_1^2 \sin^2 r_1$$

$$\text{At C, } n_1 \sin(90 - r_1) = n_2 \sin r_2$$

$$\Rightarrow n_1^2 \cos^2 r_1 = n_2^2 \sin^2 r_2$$

$$\text{At D, } n_2 \sin(90 - r_2) = n_3 \sin r_3$$

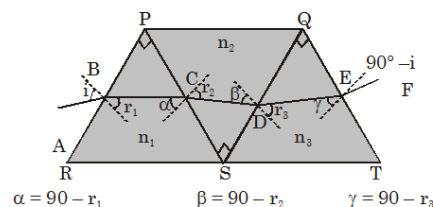
$$\Rightarrow n_2^2 \cos^2 r_2 = n_3^2 \sin^2 r_3$$

$$\text{At E, } n_3 \sin(90 - r_3) = \sin(90 - i)$$

$$\Rightarrow n_3^2 \cos^2 r_3 = \cos^2 i$$

Adding all above equations gives,

$$1 + n_2^2 = n_1^2 + n_3^2$$



8. Ans (3)

$$B = \frac{\mu_0 i r^2}{2(x^2 + r^2)^{3/2}}$$

$$= \frac{\mu_0 i r^2}{2\ell^3} \quad (i = qf)$$

$$= \frac{\mu_0 q r r^2}{2\ell^3}$$

$$= \frac{\mu_0 q f (\ell \sin 60^\circ)^2}{2\ell^3}$$

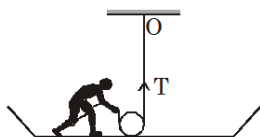
$$= \frac{\mu_0 q \omega \times 3}{16\pi \ell}$$

$$= \frac{4\pi \times 10^{-7} \times 2 \times 10^{-6} \times 5 \times 3}{16\pi \times \left(\frac{1}{4}\right)}$$

$$= 3 \times 10^{-12} \text{ T}$$

$$= 3\text{pT}$$

9. Ans (2)



$$T - (M + m)g = (M + m)a$$

$$a = \frac{50}{75} = \frac{2}{3} \text{ m/s}^2$$

$$W = \Delta k$$

$$W = k_2 - k_1$$

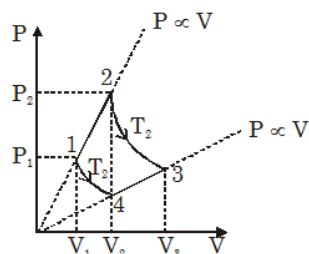
$$= \frac{1}{2}(75)(at)^2$$

$$= \frac{1}{2}(75)\left(\frac{2}{3} \times 2\right)^2$$

$$= \frac{75 \times 8}{9} = \frac{25 \times 8}{3}$$

$$\frac{3W}{100} = \frac{3 \times 25 \times 8}{100 \times 3} = 2$$

10. Ans (2)



$$\frac{P_2}{P_1} = 2$$

$$\frac{V_2}{V_1} = \frac{1}{2}$$

$$\frac{P_1}{P_2} = \frac{1}{2}$$

$$\frac{P_1}{P_2} = \frac{1}{2} \& \frac{V_1}{V_2} = \frac{1}{2}$$

$$P_1 V_1 = P_4 V_2$$

$$P_2 V_2 = P_3 V_3$$

$$\text{Now } \frac{V_3}{V_2} = \frac{P_3}{P_4} = \frac{P_2 V_2}{P_1 V_1} \times \frac{V_2}{V_3}$$

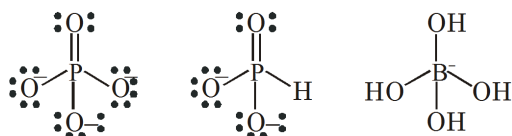
$$\frac{V_3^2}{V_2^2} = \frac{P_2}{P_1} \times \frac{V_2}{V_1}$$

$$= 2 \times 2$$

PART-2 : CHEMISTRY

SECTION-II

2. Ans (27.00)



7. Ans (10.00)



$$95\text{mg} \quad 100\text{mg}$$

$$9.5\text{mg} \quad 10\text{mg}$$

$$H = \frac{10}{10^6} \times 10^6 = 10\text{ppm}$$

8. Ans (41.50)

$$E_a = 10^{-2} \times RT + 2R$$

$$= 10^{-2} \times 8.3 \times 300 + 2 \times 8.3$$

$$= 24.9 + 16.6$$

$$= 41.5$$

9. Ans (0.46)

$$E = (-0.34 + 0.80) - \frac{0.06}{2} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2}$$

$$= 0.46 \text{ Volt}$$

10. Ans (2.50)

$$\frac{P_C \cdot V_{C,m}}{R \cdot T_C} = \frac{3}{8}$$

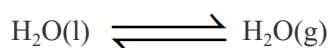
$$\frac{\frac{40}{\pi} \times \left(3 \times 4 \times \frac{4}{3} \pi r^3 \times 10^3 \frac{\text{cm}^3}{\text{mol}} \times 6 \times 10^{23}\right)}{0.08 \times 200} = \frac{3}{8}$$

$$r^3 = \frac{1}{40 \times 4 \times 4} \times 10^{-26} = \frac{1}{4^3} \times 10^{-27}$$

$$r = \frac{1}{4} \times 10^{-9} \text{ cm} = 0.25 \times 10^{-9} \text{ cm} = 2.5 \text{ pm}$$

SECTION-III

2. Ans (8)



$$\Delta_r G^\circ = \Delta G_f^\circ \text{H}_2\text{O(g)} - \Delta G_f^\circ [\text{H}_2\text{O(l)}]$$

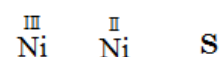
$$= -RT \ln K$$

$$= -RT \ln P_{\text{H}_2\text{O}}$$

$$= -2.303 \times \frac{8.314}{1000} \times 300 \log \frac{30}{750}$$

$$= +8 \text{ kJ (nearest integer)}$$

3. Ans (2)



$$x \quad \left(\frac{3}{4} - x\right) \quad 1.00$$

$$3x + 2 \left(\frac{3}{4} - x\right) = +2$$

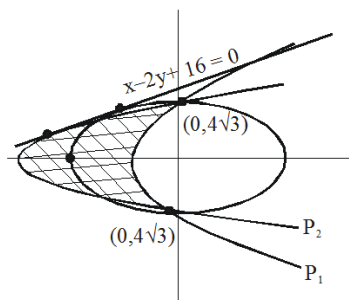
$$x = 2 - 1.5$$

$$x = 0.5$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (42.66 or 42.67)



equation of common tangent is

$$y = \frac{1}{2}x + 8 \begin{cases} m = \frac{1}{2} \\ c = 8 \end{cases}$$

Let equation of $P_1 : y^2 = 4a(x - \alpha)$

and $P_2 : y^2 = 4b(x - \beta)$

both passes through $(0, \pm 4\sqrt{3})$

$$\Rightarrow \alpha = -\frac{12}{a}, \beta = -\frac{12}{b}$$

$$\Rightarrow P_1 : y^2 = 4a \left(x + \frac{12}{a} \right)$$

$$\text{and } P_2 : y^2 = 4a \left(x + \frac{12}{b} \right)$$

equation of tangents are

$$y = m \left(x + \frac{12}{a} \right) + \frac{a}{m}, y = m \left(x + \frac{12}{b} \right) + \frac{b}{m}$$

comparing with $y = \frac{1}{2}x + 8$

$$\Rightarrow m = \frac{1}{2}, \frac{6}{a} + 2a = 8, \frac{6}{b} + 2b = 8$$

$$\Rightarrow a = 3, b = 1$$

$$\Rightarrow \alpha = -4, \beta = -12$$

equation of both the parabola are

$$y^2 = 12(x + 4), y^2 = 4(x + 12)$$

$$\begin{aligned} \Rightarrow A &= \int_{-4\sqrt{3}}^{4\sqrt{3}} \left[\left(\frac{y^2}{12} - 4 \right) - \left(\frac{y^2}{4} - 12 \right) \right] dy \\ &= \frac{128}{\sqrt{3}} \end{aligned}$$

2. Ans (2.00)

$$x(f(x)+1) = x^2 + \frac{x}{e} + \int_0^{x^2} f\left(\frac{t}{x}\right) dt \quad \forall x \in \mathbb{R}_0$$

$$\text{Let } \frac{t}{x} = z \Rightarrow \frac{1}{x} dt = dz$$

$$\Rightarrow x(f(x)+1) = x^2 + \frac{x}{e} + x \int_0^x f(z) dz$$

$$\Rightarrow f'(x) = 1 + f(x) \Rightarrow f(1) = 0$$

$$\frac{f'(x)}{1+f(x)} = 1 \quad (\because f(x) \neq -1)$$

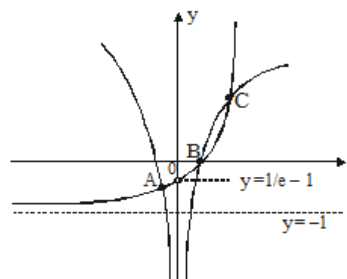
$$\ln(1+f(x)) = x + c$$

$$\because f(1) = 0 \Rightarrow c = -1$$

$$\Rightarrow f(x) = e^{x-1} - 1$$

Now, $e^{x-1} - 1 = \ln x^2$ intersects at three points as

shown in figure.



$$\Rightarrow A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$$

$$\Rightarrow n = 3, y_1 \in \left(\frac{1}{e} - 1, -1 \right), y_2 = 0$$

$$\Rightarrow [y_1] + [y_2] + n = 2$$

3. Ans (2.57)

$$\begin{aligned} |(A^{-1} - I)^2| &= |A^{-2} - 2A^{-1} + I| \\ &= \frac{|A^{-1} - 2I + A|}{|A|} \end{aligned}$$

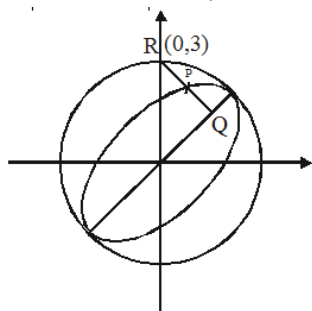
$$\because A + A^{-1} = \frac{1}{7} \begin{bmatrix} 12 & 25 & 21 \\ 20 & 23 & 77 \\ -14 & 21 & 21 \end{bmatrix} \text{ and } |A| = -7$$

$$\therefore |(A^{-1} - I)^2| = \frac{\begin{vmatrix} -\frac{2}{7} & \frac{25}{7} & 3 \\ \frac{20}{7} & \frac{9}{7} & 11 \\ -2 & 3 & 1 \end{vmatrix}}{-7} = \frac{324}{49}$$

$$\Rightarrow \text{Its positive square root} = \frac{18}{7}$$

4. **Ans (3.00)**
 $(yx^2 - e^{y^{-3}})dy = y^2 x dx$
 $\Rightarrow \frac{dx}{dy} = \frac{yx^2 - e^{y^{-3}}}{y^2 x}$
 $\Rightarrow x \frac{dx}{dy} = \frac{x^2}{y} - \frac{e^{y^{-3}}}{y^2}$
 $\Rightarrow x \frac{dx}{dy} - \frac{x^2}{y} = -\frac{e^{y^{-3}}}{y^2}$
 Put $x^2 = t$
 $\Rightarrow 2x \frac{dx}{dy} = \frac{dt}{dy}$
 $\Rightarrow \frac{1}{2} \frac{dt}{dy} - \frac{t}{y} = -\frac{e^{y^{-3}}}{y^2}$
 $\Rightarrow \frac{dt}{dy} - \frac{2}{y} \cdot t = \frac{-2}{y^2} \cdot e^{y^{-3}}$
 Integrating factor
 $I. F = \int e^{\frac{-2}{y}} dy = e^{-2 \ln y} = \frac{1}{y^2}$
 $t \cdot \frac{1}{y^2} = \int \frac{-2}{y^2} \cdot \frac{e^{y^{-3}}}{y^2} dy$
 $\frac{t}{y^2} = \frac{2}{3} e^{y^{-3}} + C$
 $\Rightarrow 3t = 2y^2 e^{y^{-3}} + Cy^2$
 $\Rightarrow \boxed{3x^2 = 2y^2 e^{y^{-3}} + Cy^2}$

5. **Ans (36.00)**
 Let length of semimajor axis of ellipse = a
 and length of semimajor axis of ellipse = b
 $a = 3$ and $RQ = \left| \frac{3.0 - 4.3}{5} \right| = \frac{12}{5}$



$$\therefore \frac{PQ}{RQ} = \frac{b}{a}$$

$$\Rightarrow \frac{b}{3} = \frac{5}{12}$$

$$\Rightarrow b = \frac{5}{4}$$

Now, distance between foci = $2ae$

$$F = 2\sqrt{a^2 - b^2} = 2\sqrt{9 - \frac{25}{16}}$$

$$\Rightarrow \frac{144F^2}{119} = 36$$

6. **Ans (0.70 or 0.71)**
 $\vec{p} \cdot \vec{q} = 0, \vec{s} \times \vec{q} = \vec{p} - \vec{s}$
 Let $\vec{s} = x\vec{p} + y\vec{q} + z(\vec{p} \times \vec{q})$
 $(x\vec{p} + y\vec{q} + z(\vec{p} \times \vec{q})) \times \vec{q} = \vec{p} - x\vec{p} - y\vec{q} - z(\vec{p} \times \vec{q})$
 $x(\vec{p} \times \vec{q}) + z[(\vec{p} \cdot \vec{q})\vec{q} - (\vec{q} \cdot \vec{q})\vec{p}] = (1 - x)\vec{p}$
 $-y\vec{q} - z(\vec{p} \times \vec{q})$
 $x(\vec{p} \times \vec{q}) - \vec{p}z = (1 - x)\vec{p} - \vec{q}y - z(\vec{p} \times \vec{q})$
 comparing $x = -z, y = 0, z = x - 1$
 $\Rightarrow x = \frac{1}{2}, y = 0, z = -\frac{1}{2}$
 $\Rightarrow \vec{s} = \frac{\vec{p} - \vec{p} \times \vec{q}}{2}$
 $\Rightarrow |\vec{s} + \vec{p} \times \vec{q}| = \left| \frac{\vec{p} - \vec{p} \times \vec{q}}{2} + \vec{p} \times \vec{q} \right|$
 $= \frac{|\vec{p} + \vec{p} \times \vec{q}|}{2} = \frac{\sqrt{|\vec{p}|^2 + |\vec{p} \times \vec{q}|^2 + 0}}{2}$
 $= \frac{1}{\sqrt{2}} = 0.707$

Alternat

Let $\vec{p} = \hat{i}, \vec{q} = \hat{j}$ and $\vec{s} = x_1 \hat{i} + x_2 \hat{j} + x_3 \hat{k}$

Now proceed

7. **Ans (20.00)**

Let $h(x) = px^2 + qx + r$ $\begin{cases} h(0) = -6 = r \\ h'(0) = \beta = q \\ h''(0) = -2\gamma = 2p \end{cases}$

$$\Rightarrow h(x) = -\gamma x^2 + \beta x - 6$$

$$\therefore h(x) \Rightarrow 4x^4 - \alpha x^3 + \gamma x^2 - \beta x + 6 = 0$$

$$\therefore A.M. \geq G.M.$$

$$\Rightarrow \frac{\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4}}{4} \geq \sqrt[4]{\frac{1.2.3.4}{x_1 x_2 x_3 x_4}}$$

$$\Rightarrow 2 \geq \left(\frac{24}{6/4} \right)^{1/4} \Rightarrow 2 \times 2$$

$$\therefore A.M. = G.M.$$

$$\therefore \frac{1}{x_1} = \frac{2}{x_2} = \frac{3}{x_3} = \frac{4}{x_4} = k \text{ (let)}$$

$$\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4} = 4k = 8 \Rightarrow k = 2$$

$$\therefore x_1 = \frac{1}{2}, x_2 = 1, x_3 = \frac{3}{2}, x_4 = 2$$

$$(2x - 1)(x - 1)(2x - 3)(x - 2) = 0$$

$$(2x^2 - 3x + 1)(2x^2 - 7x + 6) = 0$$

8. Ans (3280.00)

$$N = \underline{a} \underline{b} \underline{c} \underline{d} \underline{e}$$

Let x_r be the number of times digit r is used in

$$N. r = 0, 1, 2, 3, \dots, 9.$$

$$\text{If } N \in P \quad x_1 + x_2 + \dots + x_9 = 5 \quad \dots(i)$$

$$\text{If } N \in Q \quad x_0 + x_1 + \dots + x_9 = 5 \quad \dots(ii)$$

$$\text{From (i)} \quad n(P) = {}^{13}C_8$$

$$\text{From (ii)} \quad n(Q) = {}^{14}C_9$$

$$n(P \cap Q) = Q \{ \text{when } a = b = c = d = e \}$$

$$\text{so } n(P \cup Q) = 1287 + 2002 - 9 = 3280$$

9. Ans (106.00)

Chance that a coin is discarded

$$= \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{7}{8}$$

P(atmost two coins remains)

$$\begin{aligned} &= \left(\frac{7}{8}\right)^6 + {}^6C_5 \left(\frac{7}{8}\right)^5 \cdot \frac{1}{8} + {}^6C_4 \left(\frac{7}{8}\right)^4 \cdot \frac{1}{8^2} \\ &= \frac{2401}{8^6} [49 + 42 + 15] = \left(\frac{2401}{8^6}\right) \cdot (106) \end{aligned}$$

10. Ans (3.33 or 3.34)

$$[y] + \{y\}^3 = x, y = f(x)$$

Let $y \in [I-1, I)$, I is an integer

$$(I-1) + (f(x) - (I-1))^3 = x$$

$$f(x) = (I-1) + (x - (I-1))^{1/3}$$

Left hand derivative at $x = I$

$$\Rightarrow Lf'(I) = \lim_{h \rightarrow 0} \frac{((I-1) + (I-h-I+1)^{1/3}) - ((I-1) + 1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(1-h)^{1/3} - 1}{-h}$$

$$= \frac{1}{3}$$

$$\Rightarrow \sum_{i=0}^9 \frac{1}{3} = \frac{10}{3}$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (6)

$$x^2 - x + 1 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

where $\alpha = -\omega$, $\beta = -\omega^2$, ω is non-real cube roots of unity.

$$P_{12k+2} = 2\alpha^{12k+2} + 3\beta^{12k+2}$$

$$= 2(-\omega)^{12k+2} + 3(-\omega^2)^{12k+2}$$

$$= 2\omega^2 + 3\omega$$

$$P_{6k+1} = 2\alpha^{6k+1} + 3\beta^{6k+1}$$

$$= 2(-\omega)^{6k+1} + 3(-\omega^2)^{6k+1}$$

$$= -2\omega - 3\omega^2$$

$$\Rightarrow P_{12k+2} - P_{6k+1} = (2\omega^2 + 3\omega) - (-2\omega - 3\omega^2)$$

$$= -2 - 3 = -5$$

$$\Rightarrow \sum_{k=0}^{6n-1} |P_{12k+2} - P_{6k+1}| = \sum_{k=0}^{6n-1} 5 = 30n$$

$$\text{Now, } P_k = 2\alpha^k + 3\beta^k$$

$$\because \alpha^2 - \alpha + 1 = 0$$

$$2\alpha^{k+2} - 2\alpha^{k+1} + 2\alpha^k = 0$$

$$\text{similarly } 3\beta^{k+2} - 3\beta^{k+1} + 3\beta^k = 0$$

$$(2\alpha^{k+2} + 3\beta^{k+2}) - (2\alpha^{k+1} + 3\beta^{k+1}) + (2\alpha^k + 3\beta^k) = 0$$

$$\Rightarrow P_{k+2} + P_k = P_{k+1}$$

$$\Rightarrow \sum_{k=0}^{6n-2} (P_{k+2} + P_k) = \sum_{k=0}^{6n-2} P_{k+1}$$

$$= 2 \sum_{k=0}^{6n-2} \alpha^{k+1} + 3 \sum_{k=0}^{6n-2} \beta^{k+1}$$

$$= \frac{2\alpha \{ \alpha^{6n-1} - 1 \}}{\alpha - 1} + \frac{3 \{ \beta^{6n-1} - 1 \}}{\beta - 1}$$

$$\because x(x-1) = -1 \Rightarrow \alpha - 1 = -\frac{1}{\alpha} \quad \text{and} \quad \beta - 1 = \frac{1}{\beta}$$

$$= -2\alpha^2 \{ \alpha^{6n-1} \} - 3\beta^2 \{ \beta^{6n-1} - 1 \}$$

$$= -2\alpha^{6n+1} + 2\alpha^2 - 3\beta^{6n+1} + 3\beta^2$$

$$= -2\alpha + 2\alpha^2 - 3\beta + 3\beta^2 (\because \alpha^{6n} = \beta^{6n} = 1)$$

$$= 2(-1) + 3(-1) = -5 (\because \alpha^2 - \alpha = \beta^2 - \beta = -1)$$

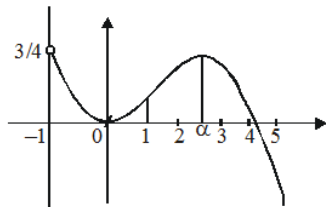
$$\Rightarrow \frac{\sum_{k=0}^{6n-1} |P_{12k+2} - P_{6k+1}|}{n \left| \sum_{k=0}^{6n-2} (P_{k+2} + P_k) \right|} = \frac{30n}{n \cdot 5} = 6$$

2. Ans (2)

$$f(x) = (1+x) \ln(1+x) - x - \frac{x^2}{4}$$

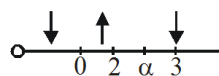
$$\lim_{x \rightarrow -1} f(x) = 0 + 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{and } f(0) = 0$$



$$f'(x) = \ln(1+x) - \frac{x}{2}$$

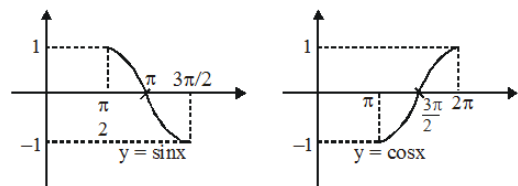
$$f'(0) = 0, f'(2) > 0, f'(3) < 0$$



$$\Rightarrow \alpha \in (2, 3)$$

3. Ans (0)

$$\text{Let } \cos I(x) = A, \cos I, \sin I(x\sqrt{3}) = B$$



$$\because A - B = C \Rightarrow \cos A \cos B + \sin A \sin B = \cos C$$

$$x\sqrt{1-x^2} + \sqrt{1-x^2} \cdot x = x\sqrt{3}$$

$$\Rightarrow x = 0, \frac{1}{2}, -\frac{1}{2}$$

Now, check $x = 0$

$$\cos I(0) - \sin I(0) = \cos I(0)$$

$$\sin I(0) = 0$$

$\pi = 0$ not possible

$$\text{check at } x = \frac{1}{2}$$

$$\cos I\left(\frac{1}{2}\right) - \sin I\left(\frac{1}{2}\right) = \cos I\left(\frac{\sqrt{3}}{2}\right)$$

$$\frac{5\pi}{3} - \frac{5\pi}{6} = \frac{11\pi}{6} \quad (\text{Not possible})$$

$$\Rightarrow x = \frac{1}{2} \text{ is not possible}$$

$$\text{Now at } x = -\frac{1}{2}$$

$$\cos I\left(-\frac{1}{2}\right) - \sin I\left(-\frac{1}{2}\right) = \cos I\left(-\frac{\sqrt{3}}{2}\right)$$

$$\frac{4\pi}{3} - \frac{7\pi}{6} = \frac{7\pi}{6} \quad \text{not possible, so number of solutions is 0.}$$

4. Ans (1)

$$x + y - z = 1$$

$$(m-1)x + (3m-7)y - 5z = -3$$

$$(m-2)x + (2m-5)y = -4$$

$\therefore$ all the three planes intersect at a line

$$\therefore \begin{vmatrix} 1 & 1 & -1 \\ m-1 & 3m-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix} = 0$$

$$\Rightarrow m = 3, -2$$

$$\text{at } m = 3 \quad x + y - z = 1$$

$$2x + 2y - 5z = -3$$

$$x + y = -4$$

system does not have any solution but at $m = -2$

$$P_1 : x + y - z = 1$$

$$P_2 : 3x + 13y + 5z = 3$$

$$P_3 : 4x + 9y = 4$$

$$\Rightarrow 5P_1 + P_2 = 2P_3$$

$\Rightarrow$ direction ratio of line of intersection are given

by

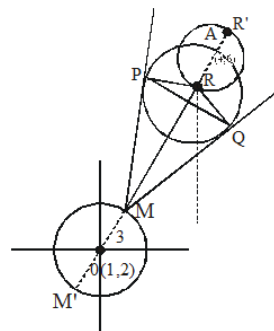
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 3 & 13 & 5 \end{vmatrix} = 18\hat{i} - 8\hat{j} + 10\hat{k}$$

$$\equiv \langle 9, -4, 5 \rangle$$

$$\Rightarrow a = 9\lambda, b = -4\lambda, c = 5\lambda$$

5. Ans (9)

Minimum value of length of diameter is MR and its maximum value is M'R'



$$\therefore MR = OA - 3 - 1 = 5 - 4 = 1 = \ell$$

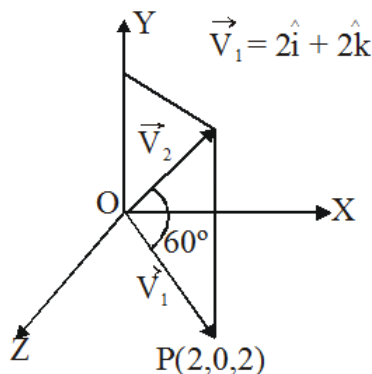
$$M'R' = MR + RR' + MM' = 1 + 2 + 6 = 9 = L$$

$$\Rightarrow \frac{L}{\ell} = 9$$

6. Ans (4)

Equation of plane perpendicular to XOZ plane

is $P : x - z = 0$ $\vec{V}_2$ is coplanar with $\vec{V}_1$ and $\hat{j}$



$$\Rightarrow \vec{V}_2 = x\vec{V}_1 + y\hat{j}$$

$$\Rightarrow \vec{V}_2 = x(2\hat{i} + 2\hat{k}) + y\hat{j}$$

$$\therefore |\vec{V}_2| = |\vec{V}_1| = 2\sqrt{2}$$

$$\therefore 4x^2 + y^2 + 4x^2 = 8 \quad \dots(1)$$

$$\text{Now, } \vec{V}_2 \cdot \vec{V}_1 = |\vec{V}_2| |\vec{V}_1| \cos \frac{\pi}{3} = 4$$

$$4 = (x(2\hat{i} + 2\hat{k}) + y\hat{j}) \cdot (2\hat{i} + 2\hat{k})$$

$$4 = 8x \Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6}$$

$$\Rightarrow \vec{V}_2 = \hat{i} + \sqrt{6}\hat{j} + \hat{k} \quad (\because \vec{V}_2 \cdot \hat{j} > 0)$$

$$\text{Now, } \vec{V}_3 = \vec{V}_1 + x\hat{i}$$

$$\therefore \vec{V}_3 \cdot \vec{V}_1 = 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4$$

$$\Rightarrow \vec{V}_3 = 2\hat{i} + 2\hat{k} - 4\hat{i}$$

$$\vec{V}_3 = -2\hat{i} + 2\hat{k}$$

$\Rightarrow$ volume of tetrahedron

$$V = \frac{1}{6} \begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix}$$

$$\Rightarrow V = \frac{1}{6} 8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4$$

7. Ans (1)

$$f(x) = \sin^2 x e^{\cot x} \quad x \in (0, \pi)$$

$$f'(x) = \sin^2 x \cdot e^{\cot x} (-\cot x \csc^2 x) + e^{\cot x} \cdot \sin 2x$$

$$= e^{\cot x} [\sin 2x - 1] = 0$$

$$\Rightarrow x = \frac{\pi}{4}$$

Now

$$f''(x) = e^{\cot x} \cdot \cos 2x \cdot 2 + (\sin 2x - 1) e^{\cot x} (-\cot x \csc^2 x)$$

$$= e^{\cot x} \left[2 \cos 2x + \frac{1 - \sin 2x}{\sin^2 x} \right]$$

$$= e^{\cot x} \left[2 \left(\frac{\cot^2 x - 1}{\cot^2 x + 1} \right) + (1 + \cot^2 x) \left(1 - \frac{2 \cot x}{1 + \cot^2 x} \right) \right]$$

Let $\cot x = t$

$$\Rightarrow 2t^2 - 2 + (1 + t^2)(1 + t^2 - 2t) = 0$$

$$(t - 1)(2t + 2) + (1 + t^2)(t - 1)^2 = 0$$

$$(t - 1)[t^3 - t^2 + 3t + 1] = 0 \quad t \in \mathbb{R}$$

$$\Rightarrow t = 1 \text{ or } t^3 - t^2 + 3t + 1 = 0$$

$$\Rightarrow x = \frac{\pi}{4}$$

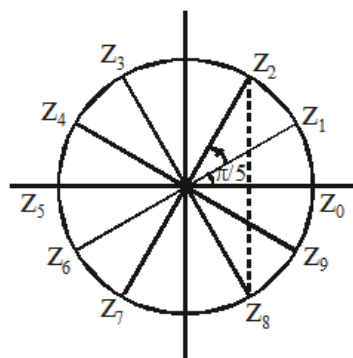
$$\Rightarrow \text{at } x = \frac{\pi}{4} f' \left(\frac{\pi}{4} \right) = f'' \left(\frac{\pi}{4} \right) = 0$$

$\Rightarrow$ Number of value of α is 1

8. Ans (8)

$$z = \cos \frac{\pi}{5} + 2i \sin \frac{\pi}{10} \cos \frac{\pi}{10}$$

$$\Rightarrow z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = e^{i\frac{\pi}{5}}$$



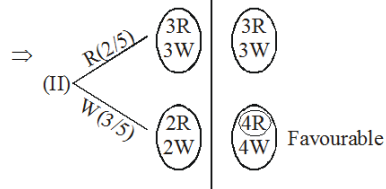
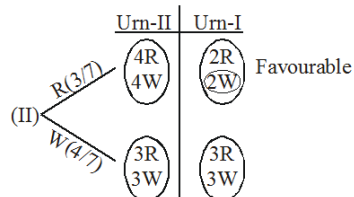
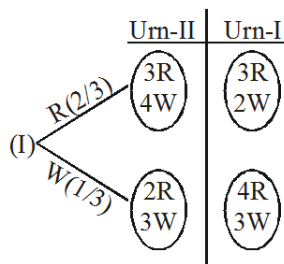
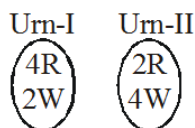
$$\therefore z_p = z_p = e^{ip\frac{\pi}{5}}, p = 0, 1, 2, 3, \dots, 9$$

$$(z_2, z_0) (z_3, z_1) (z_4, z_2) (z_5, z_3) (z_6, z_4) (z_7, z_5) (z_8, z_6)$$

$$(z_9, z_7)$$

$\Rightarrow$ 8 ordered pairs

9. Ans (7)



$\Rightarrow$ Required probability P is given by

$$P = \frac{\frac{1}{3} \times \frac{3}{5}}{\frac{1}{3} \times \frac{3}{5} + \frac{2}{3} \times \frac{3}{7}} = \frac{\frac{1}{5}}{\frac{1}{5} + \frac{2}{7}} = \frac{2 \times 5}{7 + 10} = \frac{7}{17}$$

10. Ans (6)

$$T_r = \frac{4 \cdot (2r-1)^4 + 1}{4(2r)^4 + 1} = \frac{(2(2r-1)^2 + 1)^2 - 4(2r-1)^2}{(2(2r)^2 + 1)^2 - 4(2r)^2}$$

$$= \frac{(2 \cdot (2r-1)^2 - 2 \cdot (2r-1) + 1)(2 \cdot (2r-1)^2 + 2(2r-1) + 1)}{(2(2r)^2 - 2(2r) + 1)(2(2r)^2 + 2(2r) + 1)}$$

$$= \frac{(8r^2 - 8r + 2 - 4r + 2 + 1)(8r^2 - 8r + 2 + 4r - 2 + 1)}{(8r^2 - 4r + 1)(8r^2 + 4r + 1)}$$

$$= \frac{8r^2 - 12r + 5}{8r^2 + 4r + 1}$$

$$P = T_1 \cdot T_2 \cdot \dots \cdot T_n = \frac{1}{13} \cdot \frac{13}{41} \cdot \frac{41}{85}$$

$$\dots \cdot \frac{8n^2 - 12n + 5}{8n^2 + 4n + 1} = \frac{1}{8n^2 + 4n + 1}$$

$$\Rightarrow L = \lim_{n \rightarrow \infty} \frac{n^k}{8n^2 + 4n + 1} = \text{finite}$$

$$\Rightarrow k = 2, L = \frac{1}{8}$$